

CPU Scheduling Programming Assignment - 3

Highest Response Ratio Next (HRRN)

Course: Operating Systems

September 23, 2026

1 Objective

Implement the **Highest Response Ratio Next (HRRN)** CPU scheduling algorithm in C.

HRRN is a *non-preemptive* scheduling algorithm that selects the process with the *highest response ratio*, balancing between short jobs and long-waiting jobs—thereby addressing the starvation problem inherent in Shortest Job First (SJF).

2 Background

2.1 Response Ratio Formula

For each process in the ready queue, compute:

$$\text{Response Ratio (RR)} = \frac{\text{Waiting Time} + \text{Burst Time}}{\text{Burst Time}} = 1 + \frac{\text{Waiting Time}}{\text{Burst Time}} \quad (1)$$

where

- **Waiting Time** = Current Time – Arrival Time
- **Burst Time** = Total CPU time required by the process

2.2 Selection Rule

At each scheduling decision point, among all *arrived but not-yet-executed* processes, pick the one with the **highest response ratio**. Once selected, the process runs **to completion** (non-preemptive).

2.3 Why HRRN?

- **Favors short jobs** (like SJF) $\Rightarrow$ good average turnaround time.
- **Aging effect:** longer-waiting jobs get a higher ratio $\Rightarrow$ prevents starvation of long jobs.

2.4 Metrics to Compute

- **Completion Time (CT)**
- **Turnaround Time (TAT)** = CT – Arrival Time
- **Waiting Time (WT)** = TAT – Burst Time
- **Response Time (RT)** = Start Time – Arrival Time (*For non-preemptive algorithms, RT = WT.*)

3 Requirements

Your program must:

1. Read process information from an input file.
2. Simulate HRRN scheduling.
3. Print a text-based Gantt chart.
4. Print a table with columns: PID, AT, BT, CT, TAT, WT, RT.
5. Print average TAT, WT, and RT.

4 Sample Input File

Create a file named `processes_hrrn.txt` with the following format:

```
1 Number_of_Processes
2 Process_Name Arrival_Time Burst_Time
3 Process_Name Arrival_Time Burst_Time
4 ...
5 ...
```

Listing 1: Input File Format for HRRN

Example:

```
4
P1 0 4
P2 1 6
P3 2 2
P4 3 5
```

Listing 2: Sample Input

5 Sample Walkthrough

Time	Ready Queue (with RR)	Selected	Reason
0	P1 (RR = 1.00)	P1	Only process available
4	P2: $\frac{3+6}{6} = 1.50$, P3: $\frac{2+2}{2} = 2.00$, P4: $\frac{1+5}{5} = 1.20$	P3	Highest RR = 2.00
6	P2: $\frac{5+6}{6} = 1.83$, P4: $\frac{3+5}{5} = 1.60$	P2	Highest RR = 1.83
12	P4: $\frac{9+5}{5} = 2.80$	P4	Only process left

6 Sample Output

```
Gantt Chart:
| P1 | P3 | P2 | P4 |
0    4    6    12   17

PID  AT   BT   CT   TAT  WT   RT
1    0    4    4    4    0    0
```

2	1	6	12	11	5	5
3	2	2	6	4	2	2
4	3	5	17	14	9	9
Average Turnaround Time: 8.25						
Average Waiting Time: 4.00						
Average Response Time: 4.00						

7 Deliverable

Submit a single C file: `hrrn.c`.

It must:

- Compile cleanly with `gcc`.
- Include comments explaining the selection logic.
- Handle invalid input gracefully (e.g., $n \leq 0$, negative values).

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8 Tie-Breaking Rules

If two processes have the **same response ratio**:

1. Prefer the one with the **smaller burst time**.
2. If still tied, prefer the **smaller arrival time**.
3. If still tied, prefer the **smaller PID value**.